```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, roc_auc_score, roc_curve
```

```
data = pd.read_csv("data.csv")

data.head()
```

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
0	842302	M	17.99	10.38	122.80	1001.0
1	842517	M	20.57	17.77	132.90	1326.0
2	84300903	M	19.69	21.25	130.00	1203.0
3	84348301	M	11.42	20.38	77.58	386.1
4	84358402	M	20.29	14.34	135.10	1297.0

5 rows × 33 columns

```
data = data.drop(['id','Unnamed: 32'], axis=1)

data.head()
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothnes
0	M	17.99	10.38	122.80	1001.0	(
1	M	20.57	17.77	132.90	1326.0	(
2	M	19.69	21.25	130.00	1203.0	(
3	M	11.42	20.38	77.58	386.1	(
4	M	20.29	14.34	135.10	1297.0	(

5 rows × 31 columns

```
le = LabelEncoder()
data['diagnosis'] = le.fit_transform(data['diagnosis'])

data.head()
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothnes
0	1	17.99	10.38	122.80	1001.0	(
1	1	20.57	17.77	132.90	1326.0	(
2	1	19.69	21.25	130.00	1203.0	(
3	1	11.42	20.38	77.58	386.1	(
4	1	20.29	14.34	135.10	1297.0	(

5 rows × 31 columns

```
X = data.drop('diagnosis', axis=1)
y = data['diagnosis']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
dt = DecisionTreeClassifier()

dt.fit(X_train, y_train)

dt_pred = dt.predict(X_test)
dt_prob = dt.predict_proba(X_test)[:,1]
```

```
ada = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(max_depth=1),
    n_estimators=50,
    random_state=42
)

ada.fit(X_train, y_train)

ada_pred = ada.predict(X_test)
ada_prob = ada.predict_proba(X_test)[:,1]
```

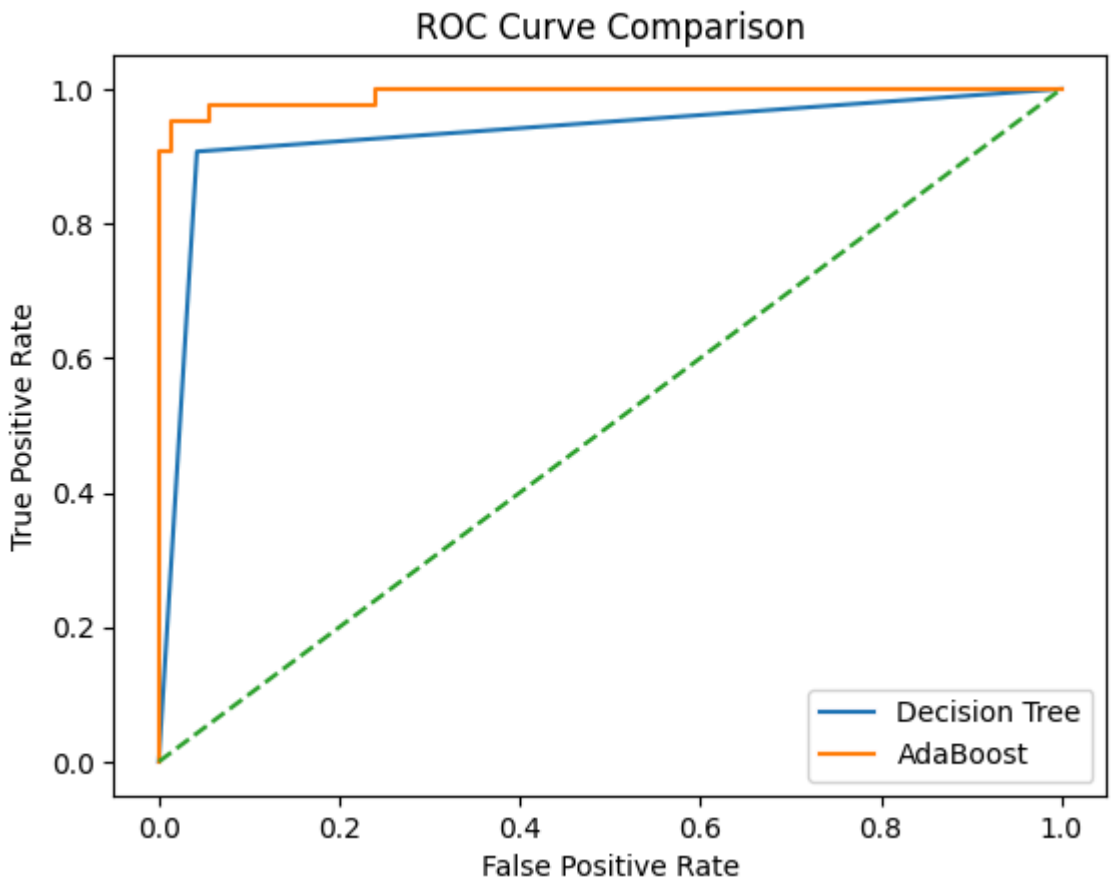
```
dt_acc = accuracy_score(y_test, dt_pred)
ada_acc = accuracy_score(y_test, ada_pred)

print("Decision Tree Accuracy:", dt_acc)
print("AdaBoost Accuracy:", ada_acc)
```

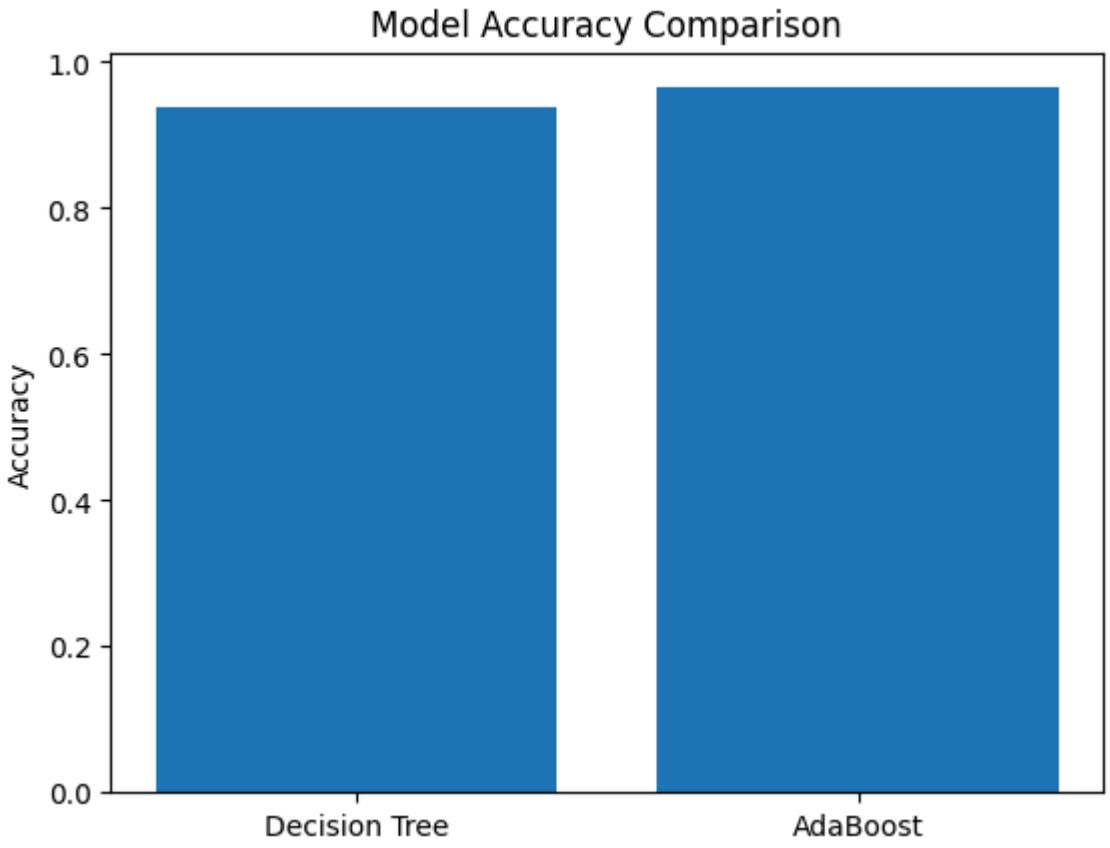
```
Decision Tree Accuracy: 0.9385964912280702
AdaBoost Accuracy: 0.9649122807017544
```

```
dt_auc = roc_auc_score(y_test, dt_prob)
ada_auc = roc_auc_score(y_test, ada_prob)

print("Decision Tree ROC-AUC:", dt_auc)
print("AdaBoost ROC-AUC:", ada_auc)
```

```
Decision Tree ROC-AUC: 0.9323616115296429
AdaBoost ROC-AUC: 0.9924664264657714
```

```
dt_fpr, dt_tpr, _ = roc_curve(y_test, dt_prob)
ada_fpr, ada_tpr, _ = roc_curve(y_test, ada_prob)

plt.figure()

plt.plot(dt_fpr, dt_tpr, label="Decision Tree")
plt.plot(ada_fpr, ada_tpr, label="AdaBoost")

plt.plot([0,1],[0,1],'--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")

plt.legend()

plt.show()
```




```
models = ['Decision Tree', 'AdaBoost']
```

```
accuracies = [dt_acc, ada_acc]

plt.bar(models, accuracies)

plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")

plt.show()
```